

Senior Helper Application - CEN3024C Observability Enhancements
Florida State College at Jacksonville

Observability Enhancements (Logger, Profiling, Validation)

To meet the deliverable requirements, observability was integrated into the SeniorHelper application using the following techniques:

Enhancement	Application Purpose	What was Used
Logging	Tracks what the application is doing at any given time (success, warnings, errors)	org.slf4j.Logger, org.slf4j.LoggerFactory
Profiling	Measures the execution time of controller operations (Create, Update, Delete)	org.slf4j.profiler.Profiler, org.slf4j.profiler.TimeInstrument
Validation	Prevents invalid API input before reaching the service layer.	@Valid and Jakarta Bean Validation annotations (ex. @NotBlank, @Email, @Size, @NotNull)

Logging – SLF4J

Logging is implemented with SLF4J across all controllers to track application activity and internal operations. Each log message includes helpful identifiers (ex. user ID, module name, appointment details) without logging sensitive data such as passwords or tokens.

Log Level	Application Purpose	Examples of What Gets Logged
WARN	Captures unusual or unexpected behavior that is not <i>fatal</i> to the application but requires attention.	- Missing or blank authentication token during user logout (Auth) - Unauthorized access attempt (Appointment)
INFO	Tracks normal application activity and successful operations. Used to understand the system flow and operations occurring in real time.	“User logged in successfully” (Auth) “Module/Appointment created / updated / deleted successfully” (Module, Appointment)
ERROR	Records failures and exception details that stop normal execution. Used for troubleshooting and root-cause analysis.	- Authentication/authorization failure (Auth) - Error during create/update/delete (Module, Appointment)

Validation - @Valid & Jakarta Bean Validation Annotations

Validation is enforced using Spring’s @Valid annotation on controller request bodies and Jakarta Bean Validation annotations (such as @NotBlank, @Email, @Size) on DTO fields. When controllers receive an incoming request, Spring validates the input and returns a ‘Bad Request’ with a helpful message if the data is missing or invalid.

Profiling – Profiler & TimeInstrument

SLF4J’s Profiler and TimeInstrument are used to determine how long service operations take within the application. Profiling starts at the beginning of controller methods and stops once the process completes with success or failure. It measures the execution time for key controller operations such as login/logout as well as appointment and module CRUD.

Sample Log Output - AuthController & AppointmentController Examples

INFO AuthController -> Login successful for user: john_senior

```
2025-11-07T12:41:28.604-05:00 INFO 42596 --- [SeniorHelper] [nio-8080-exec-1] e.f.c.s.c
.AuthController : Login successful for user: john_senior
+ Profiler [login]
|-- elapsed time [authService.login] 221.927 milliseconds.
|-- Total [login] 221.929 milliseconds.
```

WARN AuthController -> Logout attempt with missing/blank token

ERROR AuthController -> Error during logout

```
2025-11-07T12:46:37.489-05:00 WARN 42596 --- [SeniorHelper] [nio-8080-exec-3] e.f.c.s.c
.AuthController : Logout attempt with missing/blank token
2025-11-07T12:46:37.490-05:00 ERROR 42596 --- [SeniorHelper] [nio-8080-exec-3] e.f.c.s.c
.AuthController : Error during logout

org.springframework.web.server.ResponseStatusException Create breakpoint : 400 BAD_REQUEST "Missing or
blank X-AUTH-TOKEN header"
    at edu.fscj.cop3024c.seniorhelper.controller.AuthController.logout(AuthController.java:56)
> [classes/?:] <12 internal lines>
    at jakarta.servlet.http.HttpServlet.service(HttpServlet.java:590) [tomcat-embed-core-10.1.44.jar:6
> .0] <1 internal line>
    at jakarta.servlet.http.HttpServlet.service(HttpServlet.java:658) [tomcat-embed-core-10.1.44.jar:6
> .0] <33 internal lines>

+ Profiler [logout]
|-- elapsed time [authService.logout] 1655.800 microseconds.
|-- Total [logout] 1659.800 microseconds.
```

INFO AppointmentController -> Appointment updates successfully: 1

```
2025-11-07T12:48:35.736-05:00 INFO 42596 --- [SeniorHelper] [nio-8080-exec-5] e.f.c.s.c
.AppointmentController : Appointment updated successfully: 1
+ Profiler [createAppointment]
|-- elapsed time [Update Appointment] 57.441 milliseconds.
|-- Total [createAppointment] 57.448 milliseconds.
```

ERROR AppointmentController -> Error while updating appointment, Appointment not found: 0

```
2025-11-07T12:49:36.556-05:00 ERROR 42596 --- [SeniorHelper] [nio-8080-exec-7] e.f.c.s.c
.AppointmentController : Error while updating appointment Appointment not found: 0
+ Profiler [createAppointment]
|-- elapsed time [Update Appointment] 7905.700 microseconds.
|-- Total [createAppointment] 7906.800 microseconds.
```